AN R-INVARIANT OPEN SANDWICHED BETWEEN W AND A QUASI-COMPACT OPEN

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma lemma-constructing-invariant-opens (canonical tag 03LO). cited at proof lines 3873-3875 and 3890-3891. E and T are set-theoretically R-invariant.
- Lemma lemma-first-observation (canonical tag 0APA). cited at proof lines 3875-3877. Produces the quasi-compact open W' containing E.
- Lemmas 02JQ and 01R8. cited at proof lines 3883-3887. Points of the closure of T are specializations of points of T.
- Lemma morphisms-lemma-generalizations-lift-flat (03HV). cited at proof lines 3888-3890. Lifts specializations along the flat s, giving invariance of T-bar.

Source reference. Original lemma 03LO: sources/stacks-chow/tags.txt, source lines 4720–4720. Cited at proof lines 3873-3875 (E invariant) and 3890-3891 (T invariant); load-bearing.

Source reference. Original lemma 0APA: sources/stacks-chow/tags.txt, source lines 13468–13468. Cited at proof lines 3875-3877 to produce W' containing E; load-bearing.

Source reference. Original lemma 02JQ: sources/stacks-chow/tags.txt, source lines 3345–3345. Cited at proof lines 3883-3886 to reduce closedness of T-bar to specialization reachability; load-bearing.

Source reference. Original lemma 01R8: sources/stacks-chow/tags.txt, source lines 2293–2293. Cited parenthetically at proof lines 3885-3887 to justify reading T-bar as the closure of the image; load-bearing for the reading.

Source reference. Original lemma 03HV: sources/stacks-chow/tags.txt, source lines 4581–4581. Cited at proof lines 3888-3890 to lift the specialization xi' ~> xi along the flat s; load-bearing.

Lemma 0.2. *Assumptions and notation as in Lemma 0APA. There exists an R-invariant open $V \subset U$ and a quasi-compact open W' such that $W \subset V \subset W' \subset U$.*

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Proof. Set $E = t(s^{-1}(W))$. Recall that E is set-theoretically R-invariant (Lemma 03LO). By Lemma 0APA there exists a quasi-compact open W' containing E. Let $Z = U \setminus W'$ and consider $T = t(s^{-1}(Z))$. Observe that $Z \subset T$ and that $E \cap T = \emptyset$ because $s^{-1}(E) = t^{-1}(E)$ is disjoint from $s^{-1}(Z)$. Since T is the image of the closed subset $s^{-1}(Z) \subset R$ under the quasi-compact morphism $t : R \to U$ we see that any point ξ in the closure $\overline{T}$ is the specialization of a point of T, see Morphisms, Lemma 02JQ (and Morphisms, Lemma 01R8 to see that the scheme theoretic image is the closure of the image). Say $\xi' \rightsquigarrow \xi$ with $\xi' \in T$. Suppose that $r \in R$ and $s(r) = \xi$. Since s is flat we can find a specialization $r' \rightsquigarrow r$ in R such that $s(r') = \xi'$ (Morphisms, Lemma 03HV). Then $t(r') \rightsquigarrow t(r)$. We conclude that $t(r') \in T$ as T is set-theoretically invariant by Lemma 03LO. Thus $\overline{T}$ is a set-theoretically R-invariant closed subset and $V = U \setminus \overline{T}$ is the open we are looking for. It is contained in W' which finishes the proof. $\square$